

Changhwan Shin

professor

基本信息

姓名: Changhwan Shin

所属院校: 高丽大学

地区: South Korea

年龄范围: 40岁至50岁

企业经历: 有

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

纳米电子与先进工艺

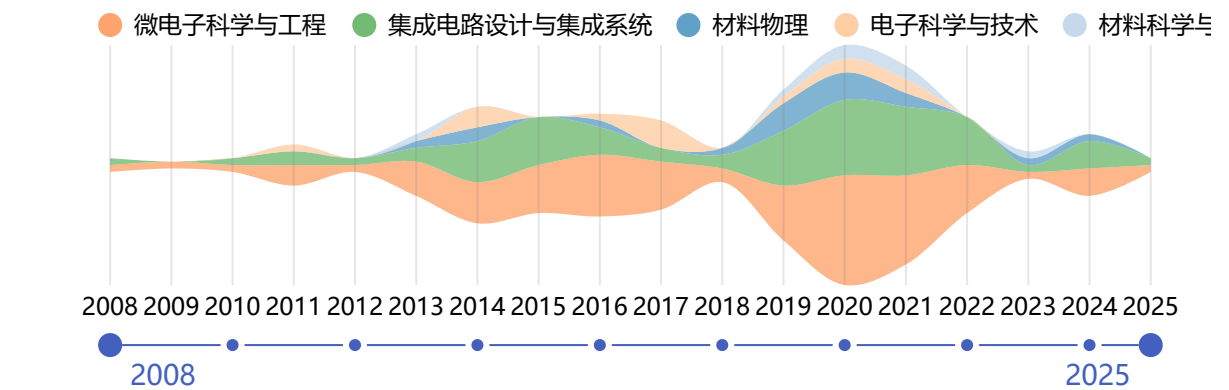
低功耗电路与能量管理

EDA工具与电路验证

个人简介

Changhwan Shin (Senior Member, IEEE), with the ORCID 0000-0001-6057-3773, is a professor at the School of Electrical Engineering, Korea University. He obtained his B.S. (Hons.) in electrical engineering from Korea University in 2006 and a Ph.D. in electrical engineering and computer sciences from the University of California at Berkeley in 2011. He has worked at Xilinx Inc. (now AMD), and held assistant/associate professor positions at the University of Seoul and Sungkyunkwan University. Since 2017, he has been a director at SK hynix. His research focuses on advanced CMOS device designs and their applications in memory/logic devices, with expertise in NAND Flash, SRAM, and more.

近年发表的论文



Taper-Angle-Induced Variation in n/p-Stacked Versus p/n-Stacked CFET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jang, Eungyo; Kim, Myongjin; Shin, Changhwan

被引用次数: 0

2025

Variation-aware analysis of buried-channel-array transistors (BCATs) in scaled DRAM: insights from 3D quasi-atomistic simulations

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Yoon, Seokchan; Lim, Jaehyuk; Shin, Changhwan

被引用次数: 0

2025

Sub-Boltzmann Switching, Hysteresis-Free Charge Modulated Negative Differential Resistance FinFET

ACS NANO

Sung, Juho; Kang, Sanghyun; Han, Donghwan; Kim, Gwon; Lim, Jaehyuk; Son, Myoungsu; Shin, Changhwan

被引用次数: 0

2024

Column Row Convolutional Neural Network: Reducing Parameters for Efficient Image Processing

NEURAL COMPUTATION

Im, Seongil; Jeong, Jae-Seung; Lee, Junseo; Shin, Changhwan; Cho, Jeong Ho; Ju, Hyunsu

被引用次数: 0

2024

Design for Variability: Counter-Doped Source/Drain Epitaxy Pockets in Gate-All-Around FET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lim, Jaehyuk; Han, Donghwan; Seon, Sangwoo; Baac, Hyoung Won; Shin, Changhwan

被引用次数: 2

2024

Row hammer-induced D0 failure improvement in sub-20 nm DRAM using an air gap

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Yoon, Jiyeong; Yoon, Seokchan; Ahn, Jinho; Shin, Changhwan

被引用次数: 0

2024

Oxygen reservoir effect of Tungsten trioxide electrode on endurance performance of ferroelectric capacitors for FeRAM applications

SCIENTIFIC REPORTS

Choi, Yejoo; Shin, Jaemin; Min, Jinhong; Moon, Seungjun; Chu, Daeyoung; Han, Donghwan; Shin, Changhwan

被引用次数: 0

2024

Buried power rail to suppress substrate leakage in complementary field effect transistor (CFET)

NANO EXPRESS

Jang, Eungyo; Lim, Jaehyuk; Shin, Changhwan

被引用次数: 0

2024

Experimental study of endurance characteristics of Al-doped HfO₂ ferroelectric capacitor

NANOTECHNOLOGY

Choi, Yejoo; Shin, Jaemin; Moon, Seungjun; Min, Jinhong; Han, Changwoo; Shin, Changhwan

被引用次数: 4

2023

Comparative Study of Novel u-Shaped SOI FinFET Against Multiple-Fin Bulk/SOI FinFET

IEEE ACCESS

Son, Myoungsu; Sung, Juho; Baac, Hyoung Won; Shin, Changhwan

被引用次数: 4

2023

Functional Encapsulating Structure for Wireless and Immediate Monitoring of the Fluid Penetration

ADVANCED FUNCTIONAL MATERIALS

Lim, Daseul; Hong, Insic; Park, Sang Uk; Chae, Jeong Woo; Lee, Seunggon; Baac, Hyoung Won; Shin, Changhwan; Lee, Jungheon; Roh, Yeonwook; Im, Chaewan; Park, Yoonseok; Lee, Geumbee; Kim, Uikyum; Koh, Je-Sung; Kang, Daeshik; Han, Seungyong; Won, Sang Min

被引用次数: 12

2022

GAN-Based Framework for Unified Estimation of Process-Induced Random Variation in FinFET

IEEE ACCESS

Park, Taeon; Kwak, Jihwan; Ahn, Hongjoon; Lee, Jinwoong; Lim, Jaehyuk; Yu, Sangho; Shin, Changhwan; Moon, Taesup

被引用次数: 2

2022

Quantitative Evaluation of Line-Edge Roughness in Various FinFET Structures: Bayesian Neural Network With Automatic Model Selection

IEEE ACCESS

Yu, Sangho; Won, Sang Min; Baac, Hyoung Won; Son, Donghee; Shin, Changhwan

被引用次数: 2

2022

Impact of Various Pulse-Bases on Charge Boost in Ferroelectric Capacitors

IEEE ELECTRON DEVICE LETTERS

Kim, Gwon; Lim, Jaehyuk; Eom, Deekjoon; Choi, Yejoo; Kim, Hyoungsub; Shin, Changhwan

被引用次数: 0

2022

Vertical Gate-All-Around Device Architecture to Improve the Device Performance for Sub-5-nm Technology

MICROMACHINES

Noh, Changwoo; Han, Changwoo; Won, Sang Min; Shin, Changhwan

被引用次数: 4

2022

Simulation Study: The Impact of Structural Variations on the Characteristics of a Buried-Channel-Array Transistor (BCAT) in DRAM

MICROMACHINES

Sun, Minjae; Baac, Hyoung Won; Shin, Changhwan

被引用次数: 2

2022

Impact of Stacking-Up and Scaling-Down Bit Cells in 3D NAND on Their Threshold Voltages

MICROMACHINES

Lee, Dongwoo; Shin, Changhwan

被引用次数: 4

2022

Study on memory characteristics of fin-shaped feedback field effect transistor

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Han, Shinick; Kim, Younghyun; Son, Donghee; Baac, Hyoung Won; Won, Sang Min; Shin, Changhwan

被引用次数: 6

2022

Design of JL-CFET (junctionless complementary field effect transistor)-based inverter for low power applications

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Lee, Sumi; Choi, Yejoo; Won, Sang Min; Son, Donghee; Baac, Hyoung Won; Shin, Changhwan

被引用次数: 4

2022

Inverter design with positive feedback field-effect transistors

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Lee, Changhoon; Han, Changwoo; Shin, Changhwan

被引用次数: 4

2022

Strain-Dependent Photoacoustic Characteristics of Free-Standing Carbon-Nanocomposite Transmitters

SENSORS

Faraz, Muhammad; Abbasi, Muhammad Awais; Son, Donghee; Shin, Changhwan; Lee, Kyu-Tae; Won, Sang Min; Baac, Hyoung Won

被引用次数: 0

2022

Experimental study of threshold voltage shift for Si:HfO₂ based ferroelectric field effect transistor

NANOTECHNOLOGY

Jung, Taehwan; Shin, Changhwan

被引用次数: 2

2021

Abruptly-Switching MoS₂-Channel Atomic-Threshold-Switching Field-Effect Transistor With AgTi/HfO₂-Based Threshold Switching Device

IEEE ACCESS

Jeong, Sojin; Han, Sangwoo; Lee, Ho-Jun; Eom, Deokjoon; Youm, Gisu; Choi, Yejoo; Moon, Seungjun; Ahn, Kyungjin; Oh, Jinju; Shin, Changhwan

被引用次数: 3

2021

Probabilistic Artificial Neural Network for Line-Edge-Roughness-Induced Random Variation in FinFET

IEEE ACCESS

Lim, Jaehyuk; Lee, Jinwoong; Shin, Changhwan

被引用次数: 11

2021

Program/Erase Scheme for Suppressing Interface Trap Generation in HfO₂-Based Ferroelectric Field Effect Transistor

IEEE ELECTRON DEVICE LETTERS

Min, Jinhong; Ronchi, Nicolo; McMitchell, Sean R. C.; O'Sullivan, Barry; Banerjee, Kaustuv; Van den Bosch, Geert; Van Houdt, Jan; Shin, Changhwan

被引用次数: 15

2021

LER-Induced Random Variation-Immune Effect of Metal-Interlayer-Semiconductor Source/Drain Structure on N-Type Ge Junction less FinFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jung, Seung-Geun; Park, Euyjin; Shin, Changhwan; Yu, Hyun-Yong

被引用次数: 3

2021

Steep-Switching Fully Depleted Silicon-on-Insulator (FDSOI) Phase-Transition Field-Effect Transistor With Optimized HfO₂/Al₂O₃-Multilayer-Based Threshold Switching Device

IEEE TRANSACTIONS ON ELECTRON DEVICES

Han, Sangwoo; Jeong, Sojin; Shin, Jaemin; Shin, Changhwan

被引用次数: 5

2021

Impact of Interface Layer on Device Characteristics of Si:HfO₂-Based FeFET's

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Jung, Taehwan; O'Sullivan, Barry J.; Ronchi, Nicolo; Linten, Dimitri; Shin, Changhwan; Van Houdt, Jan

被引用次数: 6

2021

Impact of Process-Induced Variations on Negative Capacitance Junctionless Nanowire FET

ELECTRONICS

Choi, Yejoo; Lee, Jinwoong; Lim, Jaehyuk; Moon, Seungjun; Shin, Changhwan

被引用次数: 3

2021

Impact of Rapid-Thermal-Annealing Temperature on the Polarization Characteristics of a PZT-Based Ferroelectric Capacitor

ELECTRONICS

Yu, Hanyeong; Shin, Changhwan

被引用次数: 1

2021

Prediction Model for Random Variation in FinFET Induced by Line-Edge-Roughness (LER)

ELECTRONICS

Lee, Jinwoong; Park, Taeon; Ahn, Hongjoon; Kwak, Jihwan; Moon, Taesup; Shin, Changhwan

被引用次数: 9

2021

Gate-Stack Engineering to Improve the Performance of 28 nm Low-Power High-K/Metal-Gate Device

MICROMACHINES

Park, Jeewon; Jang, Wansu; Shin, Changhwan

被引用次数: 0

2021

A Soft Pressure Sensor Array Based on a Conducting Nanomembrane

MICROMACHINES

Jung, Daekwang; Kang, Kyumin; Jung, Hyunjin; Seong, Duhwan; An, Soojung; Yoon, Jiyong; Kim, Wooseok; Shin, Mikyung; Baac, Hyoung; Won, Sangmin; Shin, Changhwan; Son, Donghee

被引用次数: 4

2021

Performance analysis and yield estimation for a negative capacitance field effect transistor-based eight-transistor static random access memory

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Hong, Yuri; Han, Changwoo; Shin, Changhwan

被引用次数: 1

2021

FBFET (feedback field-effect transistor)-based oscillator for neuromorphic computing

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Lee, Changhoon; Shin, Changhwan

被引用次数: 4

2021

Quantitative evaluation of process-induced line-edge roughness in FinFET: Bayesian regression model

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Yu, Sangho; Shin, Changhwan

被引用次数: 2

2021

Impact of depolarization electric-field and charge trapping on the coercive voltage of an Si:HfO₂-based ferroelectric capacitor

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Jung, Taehwan; Shin, Jaemin; Shin, Changhwan

被引用次数: 6

2021

Gate-induced drain leakage (GIDL) in MFMIS and MFIS negative capacitance FinFETs

CURRENT APPLIED PHYSICS

Min, Jinhong; Choe, Gihun; Shin, Changhwan

被引用次数: 7

2020

Trade-off between interfacial charge and negative capacitance effects in the Hf-Zr-Al-O/Hf_{0.5}Zr_{0.5}O₂ bilayer system

SOLID-STATE ELECTRONICS

Das, Dipjyoti; Kim, Taeho; Gaddam, Venkateswarlu; Shin, Changhwan; Jeon, Sanghun

被引用次数: 17

2020

Time-resolved electrical characteristics of ferroelectric-gated fully depleted silicon on insulator devices

SOLID-STATE ELECTRONICS

Yoon, Chankeun; Shin, Changhwan

被引用次数: 1

2020

Theoretical study of ferroelectric-gated nanoelectromechanical diode nonvolatile memory cell

SOLID-STATE ELECTRONICS

Choe, Kihun; Park, Jaesoo; Shin, Changhwan

被引用次数: 4

2020

Study of a hysteresis window of FinFET and fully-depleted silicon-on-insulator (FDSOI) MOSFET with ferroelectric capacitor

NANO CONVERGENCE

Yoon, Chankeun; Moon, Seungjun; Shin, Changhwan

被引用次数: 22

2020

Machine Learning (ML)-Based Model to Characterize the Line Edge Roughness (LER)-Induced Random Variation in FinFET

IEEE ACCESS

Lim, Jaehyuk; Shin, Changhwan

被引用次数: 16

2020

Study on Various Device Structures for Steep-Switching Silicon-on-Insulator Feedback Field-Effect Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Changhoon; Shin, Changhwan

被引用次数: 9

2020

Device Design Guideline for HfO₂-Based Ferroelectric-Gated Nanoelectromechanical System

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Yoon, Chankeun; Min, Jinhong; Shin, Jaemin; Shin, Changhwan

被引用次数: 1

2020

Energy-Delay Sensitivity Analysis of a Nanoelectromechanical Relay With the Negative Capacitance of a Ferroelectric Capacitor

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Yoon, Chankeun; Choe, Gihun; Shin, Changhwan

被引用次数: 4

2020

NCFET-Based 6-T SRAM: Yield Estimation Based on Variation-Aware Sensitivity

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Hong, Yuri; Choi, Yejoo; Shin, Changhwan

被引用次数: 8

2020

Effects of Interface Trap on Transient Negative Capacitance Effect: Phase Field Model

ELECTRONICS

Kim, Taegeon; Shin, Changhwan

被引用次数: 4

2020

MFMIS Negative Capacitance FinFET Design for Improving Drive Current

ELECTRONICS

Min, Jinhong; Shin, Changhwan

被引用次数: 9

2020

Electrical Characteristics of Nanoelectromechanical Relay with Multi-Domain HfO₂-Based Ferroelectric Materials

ELECTRONICS

Yoon, Chankeun; Shin, Changhwan

被引用次数: 1

2020

Electrical Characteristics of Bulk FinFET According to Spacer Length

ELECTRONICS

Park, Jinsu; Kim, Jaemin; Showdhury, Sanchari; Shin, Changhwan; Rhee, Hwasung; Yeo, Myung Soo; Cho, Eun-Chel; Yi, Junsin

被引用次数: 4

2020

Understanding of Polarization-Induced Threshold Voltage Shift in Ferroelectric-Gated Field Effect Transistor for Neuromorphic Applications

ELECTRONICS

Moon, Seungjun; Shin, Jaemin; Shin, Changhwan

被引用次数: 11

2020

Recent Studies on Supercapacitors with Next-Generation Structures

MICROMACHINES

Sung, Juho; Shin, Changhwan

被引用次数: 51

2020

Investigation on Threshold Voltage Adjustment of Threshold Switching Devices with HfO₂/Al₂O₃ Superlattice on Transparent ITO/Glass Substrate

MICROMACHINES

Choi, Yejoo; Shin, Jaemin; Moon, Seungjun; Shin, Changhwan

被引用次数: 4

2020

Experimental study of interface traps in MOS capacitor with Al-doped HfO₂

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Seo, Jiho; Shin, Changhwan

被引用次数: 3

2020

Device-design optimization of ferroelectric-gated vertical tunnel field-effect transistor to suppress ambipolar current

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Jung, Taehwan; Shin, Changhwan

被引用次数: 8

2020

Optimization of double metal-gate InAs/Si heterojunction nanowire TFET

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Choi, Yejoo; Hong, Yuri; Ko, Eunah; Shin, Changhwan

被引用次数: 9

2020

Compact model for PZT ferroelectric capacitors with voltage dependent switching behavior

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Wang, Chien-Wei; Ku, Hansol; Chiu, Cheng Yan; De, Sourav; Qiu, Bo-Han; Shin, Changhwan; Lu, Darsen

被引用次数: 2

2020

Study of random dopant fluctuation in PNP feedback FET

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Park, Jaesoo; Shin, Changhwan

被引用次数: 4

2020

Study of line edge roughness on various types of gate-all-around field effect transistor

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Min, Jinhong; Shin, Changhwan

被引用次数: 4

2020

Device design guideline for junctionless gate-all-around nanowire negative-capacitance FET with HfO₂-based ferroelectric gate stack

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Choi, Yejoo; Hong, Yuri; Shin, Changhwan

被引用次数: 12

2020

Understanding of Feedback Field-Effect Transistor and Its Applications

APPLIED SCIENCES-BASEL

Lee, Changhoon; Sung, Juho; Shin, Changhwan

被引用次数: 21

2020

Experimental observation of zero DIBL in short-channel hysteresis-free ferroelectric-gated FinFET

SOLID-STATE ELECTRONICS

Shin, Jaemin; Shin, Changhwan

被引用次数: 6

2019

Impact of negative capacitance on the energy-delay property of an electromechanical relay

JAPANESE JOURNAL OF APPLIED PHYSICS

Choe, Kihun; Shin, Changhwan

被引用次数: 4

2019

Ultra-thick semi-crystalline photoactive donor polymer for efficient indoor organic photovoltaics

NANO ENERGY

Shin, Sang-Chul; Koh, Chang Woo; Vincent, Premkumar; Goo, Ji Soo; Bae, Jin-Hyuk; Lee, Jae-Joon; Shin, Changhwan; Kim, Hyeok; Woo, Han Young; Shim, Jae Won

被引用次数: 85

2019

Rhenium Diselenide (ReSe₂) Near-Infrared Photodetector: Performance Enhancement by Selective p-Doping Technique

ADVANCED SCIENCE

Kim, Jinok; Heo, Keun; Kang, Dong-Ho; Shin, Changhwan; Lee, Sungjoo; Yu, Hyun-Yong; Park, Jin-Hong

被引用次数: 34

2019

Polarity control in a single transition metal dichalcogenide (TMD) transistor for homogeneous complementary logic circuits

NANOSCALE

Shim, Jaewoo; Jang, Sung Woon; Lim, Ji-Hye; Kim, Hyeongjun; Kang, Dong-Ho; Kim, Kwan-Ho; Seo, Seunghwan; Heo, Keun; Shin, Changhwan; Yu, Hyun-Yong; Lee, Sungjoo; Ko, Dae-Hong; Park, Jin-Hong

被引用次数: 21

2019

Negative quantum capacitance effect from Bi₂Te_{1.5}Se_{1.5} with frequency dependent capacitance of polyvinyl alcohol (PVA) film in MOS structure

APPLIED SURFACE SCIENCE

Choi, Hyunwoo; Park, June; Shim, Jae Won; Shin, Changhwan

被引用次数: 5

2019

Comprehensive study of high pressure annealing on the ferroelectric properties of Hf_{0.5}Zr_{0.5}O₂ thin films

NANOTECHNOLOGY

Oh, Changyong; Tewari, Amit; Kim, Kyungkwan; Kumar, Ulayil Sajesh; Shin, Changhwan; Ahn, Minho; Jeon, Sanghun

被引用次数: 8

2019

Precise control of nanoscale spacing between electrodes using different natured self-assembled monolayers

NANOTECHNOLOGY

You, Young-Jun; Kim, Yongjin; Cheun, Hyeunseok; Shin, Changhwan; Kim, Do Young; Lee, Jae Hak; Song, Joon Yub; Lee, Jae Woo; Lee, Sae Youn; Ju, Byeong-Kwon; Shim, Jae Won

被引用次数: 0

2019

Negative Capacitance Transistor with Two-Dimensional Channel Material (Molybdenum disulfide, MoS₂)

PHYSICA STATUS SOLIDI A-APPLICATIONS AND MATERIALS SCIENCE

Choi, Hyunwoo; Shin, Changhwan

被引用次数: 9

2019

Influence of High-Pressure Annealing on Memory Properties of Hf_{0.5}Zr_{0.5}O₂ Based 1T-FeRAM

IEEE ELECTRON DEVICE LETTERS

Yoon, Jae Seok; Tewari, Amit; Shin, Changhwan; Jeon, Sanghun

被引用次数: 13

2019

Steep Slope Silicon-On-Insulator Feedback Field-Effect Transistor: Design and Performance Analysis

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Changhoon; Ko, Eunah; Shin, Changhwan

被引用次数: 38

2019

Ferroelectric-Gated Nanoelectromechanical Nonvolatile Memory Cell

IEEE TRANSACTIONS ON ELECTRON DEVICES

Choe, Kihun; Shin, Changhwan

被引用次数: 9

2019

Tunnel Field-Effect Transistor With Segmented Channel

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Park, Jaesoo; Shin, Changhwan

被引用次数: 2

2019

External Resistor-Free Gate Configuration Phase Transition FDSOI MOSFET

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Shin, Jaemin; Shin, Changhwan

被引用次数: 5

2019

Study of work-function variation in stacked multiple-channel-structure device

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Park, Jaesoo; Shin, Changhwan

被引用次数: 1

2019

Experimental understanding of polarization switching in PZT ferroelectric capacitor

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Shin, Changhwan

被引用次数: 7

2019

DIBL improvement in hysteresis-free and ferroelectric-gated FinFETs

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Shin, Jaemin; Shin, Changhwan

被引用次数: 1

2019

DIBL enhancement in ferroelectric-gated FinFET

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Cho, Hyungki; Shin, Changhwan

被引用次数: 9

2019

Super steep-switching ($SS \approx 2$ mV/decade) phase-FinFET with Pb(Zr_{0.52}Ti_{0.48})O₃ threshold switching device

APPLIED PHYSICS LETTERS

Shin, Jaemin; Ko, Eunah; Park, June; Kim, Seung-Geun; Lee, Jae Woo; Yu, Hyun-Yong; Shin, Changhwan

被引用次数: 9

2018

Steep switching devices for low power applications: negative differential capacitance/resistance field effect transistors

NANO CONVERGENCE

Ko, Eunah; Shin, Jaemin; Shin, Changhwan

被引用次数: 69

2018

Impact of the Metal-Gate Material Properties in FinFET (Versus FD-SOI MOSFET) on High- κ /Metal Gate Work-Function Variation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Nam, Hyohyun; Shin, Changhwan; Park, Jung-Dong

被引用次数: 10

2018

Tapered Coating for Nano-electro-mechanical (NEM) Relay to Improve Energy-delay Product

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Choe, Kihun; Shin, Changhwan

被引用次数: 0

2018

Effective drive current in steep slope FinFET (vs. conventional FinFET)

APPLIED PHYSICS LETTERS

Ko, Eunah; Shin, Changhwan

被引用次数: 5

2017

Measurement of the quantum capacitance from two-dimensional surface state of a topological insulator at room temperature

APPLIED SURFACE SCIENCE

Choi, Hyunwoo; Kim, Tae Geun; Shin, Changhwan

被引用次数: 7

2017

Impact of Source/Drain Metal Work Function on the Electrical Characteristics of Anatase TiO₂-Based Thin Film Transistors

ECS JOURNAL OF SOLID STATE SCIENCE AND TECHNOLOGY

Choi, Hyunwoo; Shin, Jaemin; Shin, Changhwan

被引用次数: 19

2017

Current-Voltage Model for Negative Capacitance Field-Effect Transistors

IEEE ELECTRON DEVICE LETTERS

Lee, Hyunjae; Yoon, Youngki; Shin, Changhwan

被引用次数: 35

2017

Negative Capacitance FinFET With Sub-20-mV/decade Subthreshold Slope and Minimal Hysteresis of 0.48 V

IEEE ELECTRON DEVICE LETTERS

Ko, Eunah; Lee, Jae Woo; Shin, Changhwan

被引用次数: 109

2017

Transconductance Amplification by the Negative Capacitance in Ferroelectric-Gated P3HT Thin-Film Transistor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jo, Jaesung; Kim, Min Gee; Lee, Hyunjae; Choi, Hyunwoo; Shin, Changhwan

被引用次数: 11

2017

Adjusting the Operating Voltage of an Nanoelectromechanical Relay Using Negative Capacitance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Choe, Kihun; Shin, Changhwan

被引用次数: 30

2017

Impact of Interface Traps and Surface Roughness on the Device Performance of Stacked-Nanowire FETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Jinyoung; Shin, Changhwan

被引用次数: 6

2017

Impact of Equivalent Oxide Thickness on Threshold Voltage Variation Induced by Work-Function Variation in Multigate Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Youngtaek; Shin, Changhwan

被引用次数: 31

2017

Silver Nanowire/Colorless-Polyimide Composite Electrode: Application in Flexible and Transparent Resistive Switching Memory

SCIENTIFIC REPORTS

Yeom, Seung-Won; You, Banseok; Cho, Karam; Jung, Hyun Young; Park, Junsu; Shin, Changhwan; Ju, Byeong-Kwon; Kim, Jong-Woong

被引用次数: 27

2017

Experimental evidence of negative quantum capacitance in topological insulator for sub-60-mV/decade steep switching device

APPLIED PHYSICS LETTERS

Choi, H.; Lee, H.; Park, J.; Yu, H. -Y.; Kim, T. G.; Shin, C.

被引用次数: 5

2016

Threshold voltage variation-immune FinFET design with metal-interlayer-semiconductor source/drain structure

CURRENT APPLIED PHYSICS

Shin, Changho; Kim, Jeong-Kyu; Shin, Changhwan; Kim, Jong-Kook; Yu, Hyun-Yong

被引用次数: 3

2016

Effect of Metal Nitride on Contact Resistivity of Metal-Interlayer-Ge Source/Drain in Sub-10-nm n-Type Ge FinFET

IEEE ELECTRON DEVICE LETTERS

Ahn, Juhan; Kim, Jeong-Kyu; Kim, Sun-Woo; Kim, Gwang-Sik; Shin, Changhwan; Kim, Jong-Kook; Cho, Byung Jin; Yu, Hyun-Yong

被引用次数: 3

2016

Negative Capacitance Field Effect Transistor With Hysteresis-Free Sub-60-mV/Decade Switching

IEEE ELECTRON DEVICE LETTERS

Jo, Jaesung; Shin, Changhwan

被引用次数: 188

2016

3-D Quasi-Atomistic Model for Line Edge Roughness in Nonplanar MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Oh, Sangheon; Shin, Changhwan

被引用次数: 16

2016

Vertical Tunnel FET: Design Optimization With Triple Metal-Gate Layers

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ko, Eunah; Lee, Hyunjae; Park, Jung-Dong; Shin, Changhwan

被引用次数: 88

2016

Design for Variation-Immunity in Sub-10-nm Stacked-Nanowire FETs to Suppress LER-induced Random Variations

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Jinyoung; Lee, Hyunjae; Oh, Sangheon; Shin, Changhwan

被引用次数: 13

2016

Study of Random Variation in Germanium-Source Vertical Tunnel FET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Hyunjae; Park, Jung-Dong; Shin, Changhwan

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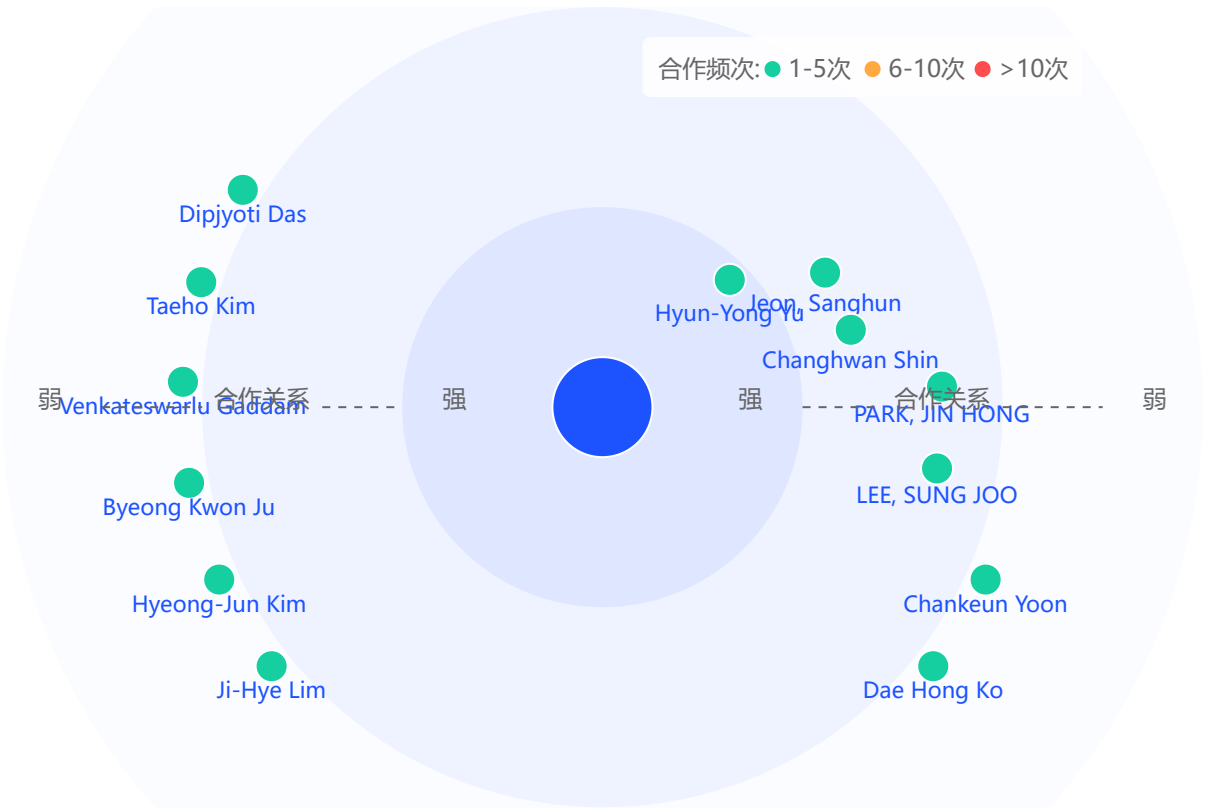
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学术合作



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